Statistical Hypothesis Testing Notes

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1. Chi-Square Goodness of Fit Test

Question: In the 2010 census of a city, the weight distribution of individuals in a small city were found to be the following.

Weight	2010 (%)	2020 Observed
<50 kg	20%	140
50–75 kg	30%	160
>75 kg	50%	200

Expected Frequencies:
$0.2 \times 500 = 100$
$0.3 \times 500 = 150$
$0.5 \times 500 = 250$

H_0: The data meets expectation.
H_1: The data does not meet expectation.

$\alpha = 0.05$, Confidence Interval = 95%

Degrees of Freedom: $dof = k - 1 = 3 - 1 = 2$

Critical Value: 5.99

Decision Rule: If $\chi^2 > 5.99$ reject H_0, else fail to reject H_0.

Formula:
$\chi^2 = \Sigma ((O-E)^2 / E)$

$\chi^2 = (40^2/100) + (10^2/150) + ((-50)^2/250)$
$= 1600/100 + 100/150 + 2500/250$
$= 16 + 0.66 + 10$
$= 26.66$

Conclusion: $26.66 > 5.99$, therefore Reject H_0.

2. One-Way ANOVA

Question: A doctor wants to test a new medicine to reduce headache. Participants are split into 3 conditions (15 mg, 30 mg, 45 mg).

15 mg	30 mg	45 mg
9	7	4
8	6	3
7	6	2
8	7	3
8	8	4
9	7	3
8	6	2

H_0: $\mu 15 = \mu 30 = \mu 45$
H_1: At least one mean is not equal.

$\alpha=0.05$, CI=95%

N=21, a=3, n=7
dof between=$a-1=2$
dof within=$N-a=18$
dof total=$N-1=20$

Critical Value = 3.5546

Decision Rule: If F > 3.5546 reject H_0 else fail to reject H_0.

SS Between Formula:
$SS_{between} = \Sigma(Ai^2)/n - T^2/N$

$=(57^2+47^2+21^2)/7 - (57+47+21)^2/21$
15 mg total=57
30 mg total=47
45 mg total=21
SS Between=98.67

SS Within Formula:
$SS_{within} = \Sigma Y^2 - \Sigma(Ai^2)/n$
$\Sigma Y^2=853$
SSwithin=10.29

Source	SS	DOF	MS
Between	98.67	2	49.34
Within	10.29	18	0.54
Total	108.95	20	—

F Formula:
F = MSbetween / MSwithin = 49.34 / 0.54 = 86.56

Conclusion: 86.56 > 3.5546, therefore Reject H_0.

3. One Sample Z-Test

Question: Population mean=168 cm, population SD=3.9, sample size=36, sample mean=169.5 cm.

H_0: $\mu=168$
H_1: $\mu\neq168$

Two-tailed test
$\alpha=0.05$, Confidence Interval=95%

Decision Boundary: -1.96 to 1.96

Decision Rule: If Z lies between -1.96 and 1.96, fail to reject H_0; otherwise reject H_0.

Z Formula:
$Z=(\bar{x}-\mu)/(\sigma/\sqrt{n})$

$=(169.5-168)/(3.9/\sqrt{36})$
$=1.5/0.65$
$=2.31$

Conclusion: $2.31 > 1.96$, therefore Reject H_0.